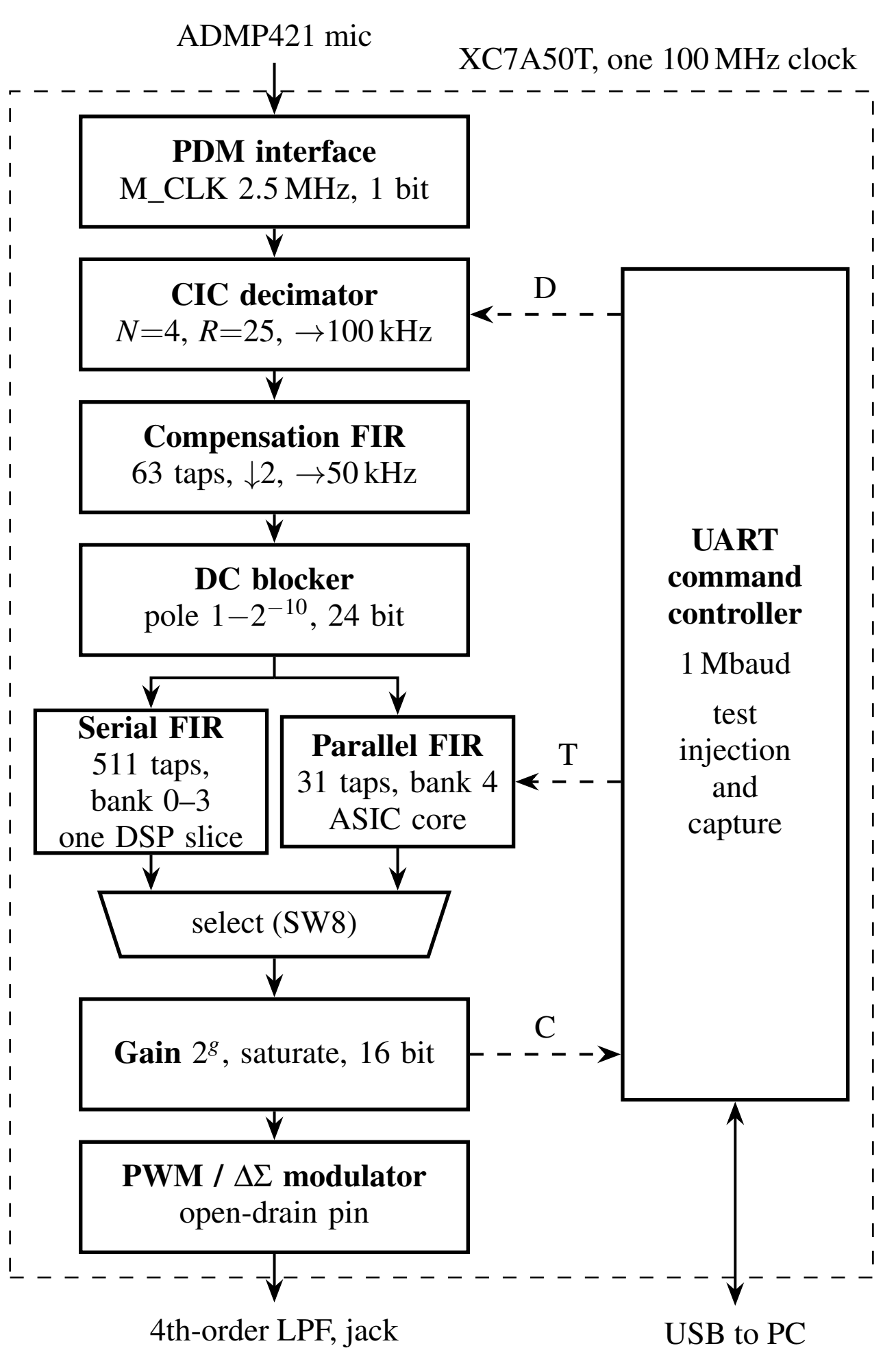

ADMP421 mic
XC7A50T, one 100 MHz clock
PDM interface
M_CLK 2.5 MHz, 1 bit
CIC decimator
N=4, R=25, →100 kHz
Compensation FIR
63 taps, ↓2, →50 kHz
DC blocker
pole 1−2−10, 24 bit
Serial FIR
511 taps,
bank 0–3
one DSP slice
Parallel FIR
31 taps, bank 4
ASIC core
select (SW8)
Gain 2g, saturate, 16 bit
PWM / ΔΣ modulator
open-drain pin
D
T
C
UART
command
controller
1 Mbaud
test
injection
and
capture
4th-order LPF, jack
USB to PC